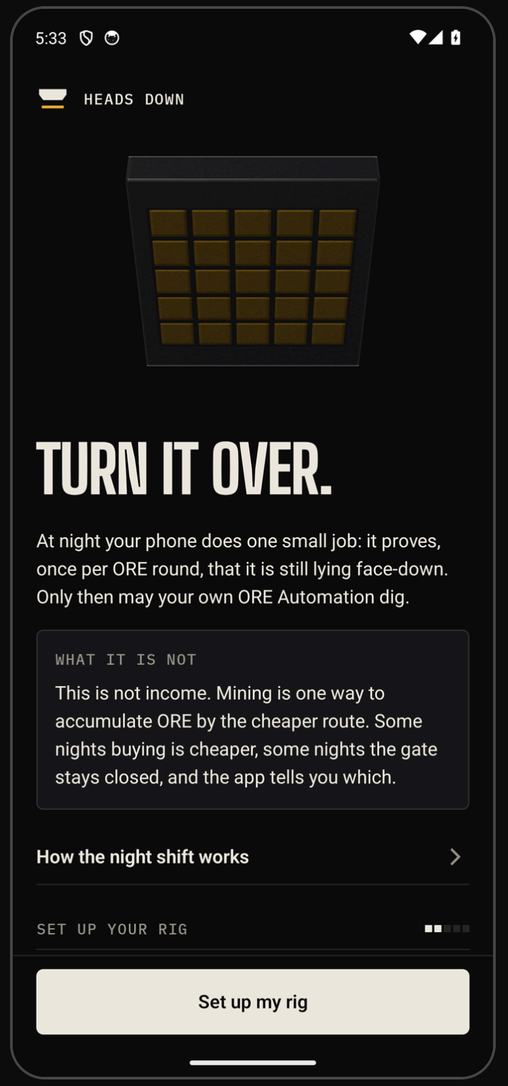
HEADS DOWN

YOUR PHONE'S NIGHT SHIFT.

Face-down, it digs ORE. Pick it up, the rig goes cold.

Any Android. Verified Seeker tier. Powered by ORE.

CLOCK IN · THE SOLANA MOBILE HACKATHON
ON MAINNET SINCE 10 OCTOBER 2026 · FIRST SHIFT DUG THAT EVENING, FROM ONE REDMI 14C



The app's first screen (Android 14 emulator)

SOURCE README.md · deploy/receipts/mainnet/20261010T184116Z-fresh-d67a1a40c2a0.json · docs/MAINNET.md (the first shift)

Most of these phones sit idle. Heads Down gives the idle one a job.

121,035

Seeker Genesis Tokens minted

the on-chain group size when read

118.9k of 121k

.skr IDs dormant, approximately

SeekerTracker, a third party

9,000+

daily active users, Seeker Season 2

June 2026, a third party

Three figures from different sources and dates. None is divided by another.

SOURCE crates/sgt-verify/README.md (group size) · docs/research/redteam.md (.skr IDs) · docs/research/retention-pmf.md (daily users)

THE GAP

ORE advertised mining on Seeker and never shipped a native app.

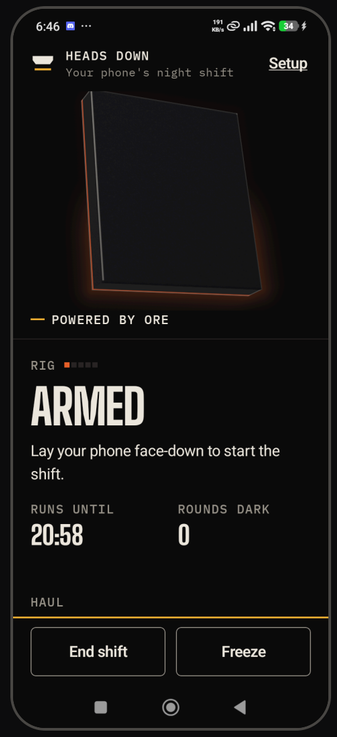
2025	26 Sep 2025	6 Nov 2025	20 Aug 2026	29 Sep 2026	10 Oct 2026
ORE's landing card: “Proof of mobile … Coming soon”	ORE adds `claim_seeker`	ORE removes it	ore.com becomes a PWA	165 to 172 miners per round on mainnet	**heads_down on mainnet**

The store's autominers (RefinORE, Orestack, Oreminer) run with the phone's app closed.

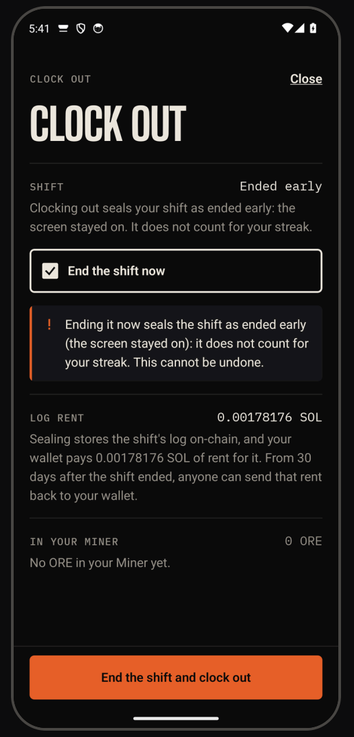
SOURCE docs/research/skr-and-ore.md · docs/ORE.md §1 (miners per round) · docs/research/redteam.md (store autominers) · deploy/receipts/mainnet/

One wallet approval at bedtime. The phone does the rest.

1	Tap Clock in, in the app or on its Quick Settings tile.	On the Redmi
2	One wallet approval funds a capped shift in your own ORE Automation and arms the rig.	On mainnet
3	Face-down, the phone signs a heartbeat every ORE round, about a minute.	On mainnet
4	A dig happens only when the on-chain price gate is open.	On mainnet
5	Pick it up: 10 s of cooling, then cold. Unlock: cold at once.	On mainnet
6	At your alarm, the morning haul reveal replays the night from chain data.	Not on a phone yet
7	Clock out: seal the shift. Take your SOL back whenever you like.	On mainnet



Armed, on the Redmi



Clock-out (emulator)

FROM THE REDMI 14C 10 Oct 2026, on mainnet, under the demo ceiling: clock-in 4DvP…aQht, clock-out 3Vfb…uTUm, SOL back 46NX…DcFF: 0.00502704 SOL, to the lamport.

SOURCE docs/MAINNET.md (the first shift, every transaction) · android/README.md · scripts/devstack/emulator-smoke.sh (the clock-out picture: emulator, Solana Mobile's test wallet)

One object on every screen: a dark slab, lit from underneath.

THE IDEA

The app works while you cannot see it. So its mark is the face you never look at: light from under a slab.

THE OBJECT

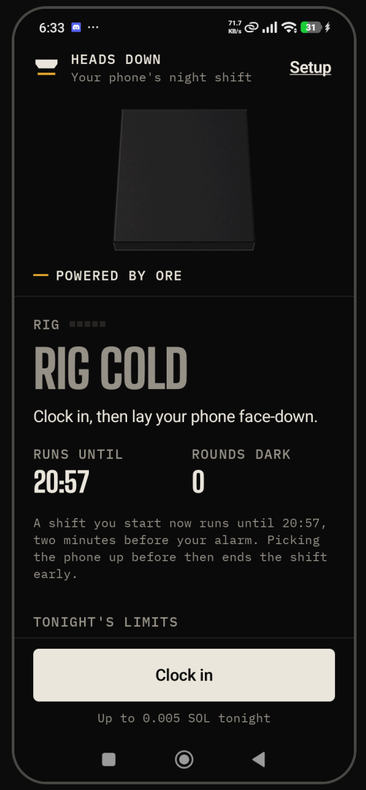
One shader draws the slab. It stays level with the real world through the accelerometer: tilt the phone, or drag, to look underneath.

THE COLOUR

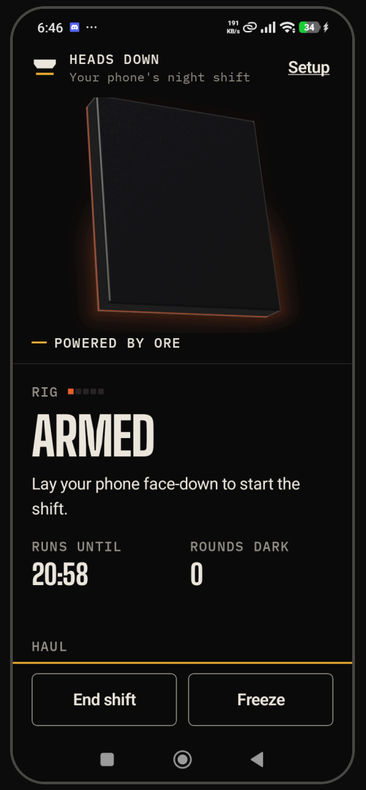
One accent at two temperatures. Ember while the rig is armed or cooling; gold only for a hot rig and for ORE.

THE RULES

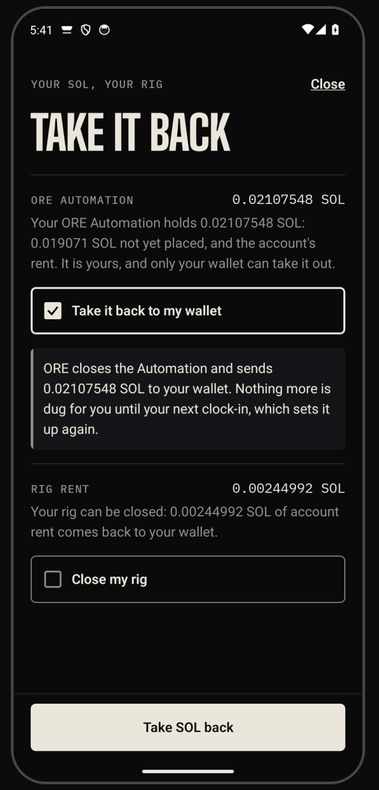
One action, in the thumb zone. The most a shift can place beside it, before the wallet opens. Nothing moves by itself.



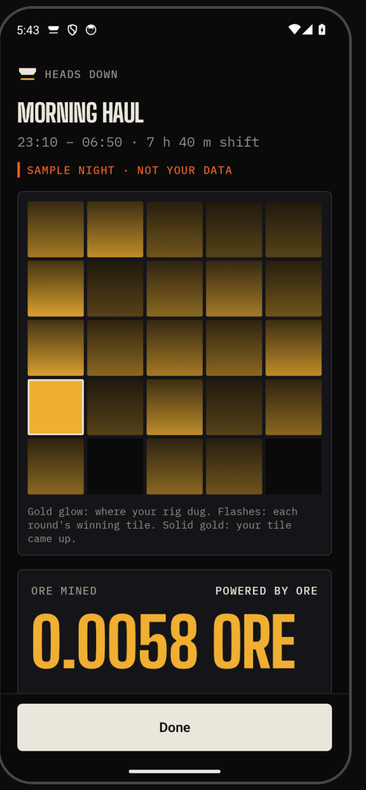
Cold, on the Redmi



Armed, on the Redmi



Take it back (emulator)



The reveal, sample night (emulator)

The first two: the app on the Redmi 14C, on mainnet, 11 Oct 2026, where this build clocked in three times and dug ten rounds. The last two: the same build on an Android 14 emulator against the local fork of mainnet; the reveal shows the sample night the app offers before a first haul, and says so on screen.

SOURCE android/core/design/ (tokens, type, components) · android/app/src/main/kotlin/xyz/headsdown/ui/slab/ (the slab) · docs/MAINNET.md (the third evening: the phone) · scripts/devstack/emulator-smoke.sh (the emulator)

The executor signs ORE's deploy only for a fresh heartbeat from the phone.

THE PHONE
Keystore key
P-256. It signs a heartbeat each ORE round and never leaves the phone.

→

ANYONE
Crank
Carries the heartbeat to the chain. It cannot forge one.

→

SOLANA
secp256r1 precompile
Verifies the signature in the same transaction.

→

HEADS_DOWN
Executor PDA
Signs only if this rig's key signed this round. It picks amount and squares.

→

ORE
deploy
From your own Automation. Your wallet funds it; ORE owns it.

NO FRESH HEARTBEAT, NO DEPLOY.

171 program tests, 138 of them on a fork of live mainnet ORE

A DIG, FROM THE REDMI
25nB…zovY

A PICKUP'S BREAK
63FZ…yqVb

A REPLAYED HEARTBEAT, REFUSED
4EJv…PUpM

OPEN TODAY One key, the founder's, can upgrade the program: no multisig, no delay. No third-party audit. The demo build digs under a raised price ceiling.

SOURCE programs/heads-down/INTERFACE.md (dig) · spikes/ore-executor/README.md · programs/heads-down/README.md (tests) · docs/THREAT_MODEL.md (K5) · docs/DEPLOY.md §10.7

MEASURED

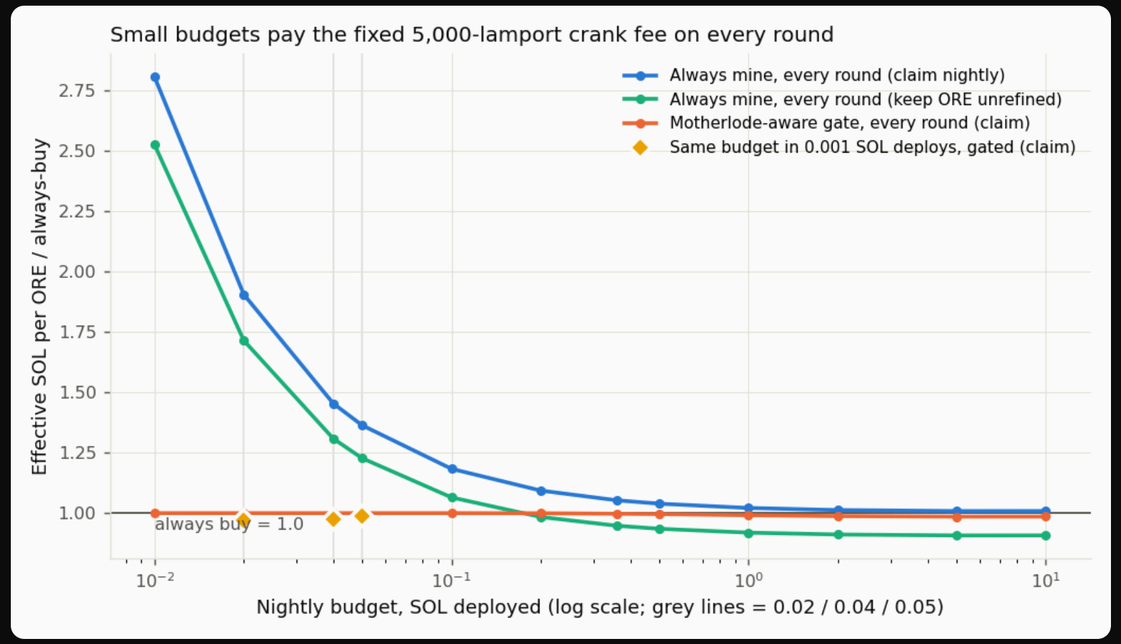
The core path is measured on the live ORE binary, not estimated.

WHAT	VALUE
Compute for a dig	39,414 CU with one rig · 67,421 with two
Rigs per dig transaction	legacy 2 · v0 with a lookup table 5 · v1 11
secp256r1 check	477 CU of introspection · 5,000 lamports a signature
Crank cost per rig-dig	fork: about 6,040 lamports in a batch of five · mainnet, one rig alone: 11,028 to 11,082 · 7,000 reimbursed
Seeker check	about 1,900 CU
The binary on mainnet	190,048 bytes, SBPF v3 · on-chain bytes equal the build

Rows 1 to 5: LiteSVM on a fork of live mainnet ORE, and a local validator; the mainnet figure in row 4 is from the first shift's four digs at a priority price of 20,000. Row 6: the deploy receipt.

SOURCE programs/heads-down/README.md (Measurements) · crank/README.md · docs/MAINNET.md (What it cost) · spikes/secp256r1/README.md · crates/sgt-verify/README.md · deploy/receipts/mainnet/

Spread thin, mining costs more than buying. Gated chunks came out slightly cheaper in a backtest.



The backtest's figure, unchanged. 1.0 is always buying.

58,801 REAL ORE ROUNDS, 47 NIGHTS	AGAINST BUYING
Dig every round	+36% to +90%
Dig 0.001 SOL chunks when the gate opens, buy the rest	−1.4% to −3.0%

THE FEE The backtest charged 5,000 lamports a dug round. Mainnet charges 10,000; it has not been re-run at that fee.

THE DEMO ORE's cost was 0.69 to 0.70 SOL per ORE during the first shift and 0.78 on 11 October: above the default plan of 0.53, so a default build does not dig. The demo build plans 1.0, under a ceiling of 1.2.

SOURCE ml/forecaster/RESULTS.md (TL;DR) · ml/forecaster/orelib.py (the backtest's fee) · deploy/receipts/mainnet/20261010T184715Z-init.json (the fee on mainnet) · docs/MAINNET.md and the crank's

The AI runs on the phone and can only tighten the limits the wallet signed.

Cost Forecaster

`Built, advisory`

9.2% lower error one hour ahead than “same as now”. It lost to the on-chain rule, so it never decides.

Pickup detection

`Built`

Accelerometer only, with hysteresis. The Redmi 14C has no gyroscope.

Pickup or bump classifier

`Built, synthetic data`

Trained on synthetic motion. No recording from a real phone yet, so its scores say nothing about real nights.

Shift Planner

`Built, simulated users`

Proposes the shift window from the app's own log.

A phone-signed plan above the wallet's caps fails on-chain: `PlanExceedsCaps`. Sensor data never leaves the phone.

SOURCE ml/forecaster/RESULTS.md §5 · ml/forecaster/MODEL_CARD.md · ml/foreman/README.md · android/feature/shift/…/FaceDownDetector.kt · programs/heads-down/README.md (shifts suite) · docs/PRIVACY.md §1

TWO TIERS

Guest rigs run on any Android. Seeker rigs are verified in-program by their SGT.

GUEST RIG

On mainnet

Any Android with a mobile wallet.

On the Redmi 14C: a rig key in the TEE that the live registrar attested, registered on mainnet as a guest with attestation level 1, and a shift signed in Jupiter's wallet.

SEEKER RIG

Built, on a fork only

One verified rig per SGT mint, anchored on Token-2022 group membership.

No Seeker device: tested on a fork of mainnet, on real SGTs #20 and #121,035.

A mint forged with the real Token-2022 program is rejected: `MissingGroupMember`.

SOURCE crates/sgt-verify/README.md · programs/heads-down/README.md (seeker suite) · docs/MAINNET.md (the first shift: RigRegistered, tier 0, attestation level 1)

Native Kotlin and Compose, on system surfaces a web app cannot reach.

- Quick Settings tile or the app, then one wallet approval (Mobile Wallet Adapter 2.2.0)
- One clock-in transaction: ORE automate, register, caps, arm
- Foreground service. Killed by the OS means cold, nothing spent
- HyperOS keep-alive setup and a “killed last night” check
- Android 16 Live Update notification, with a 14/15 fallback
- A clock-in the wallet sent is found on-chain if the app never heard so

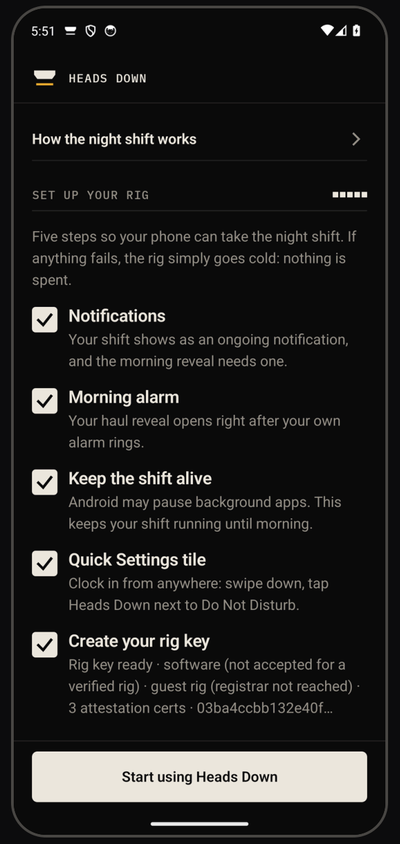
1,330

JVM unit tests passed; 5 more are skipped

Redmi 14C

Android 16, HyperOS 3, accelerometer only. On 10 and 11 Oct 2026: setup, a TEE rig key, attestation, and five short shifts on mainnet, three of them on the redesigned build.

On mainnet



Setup: five real steps (emulator)

SOURCE android/README.md · android/INTERFACE-NOTES.md §4 (one legacy packet) · docs/MAINNET.md · docs/SECURITY_REVIEW.md §5 (1,330 tests)

SKR is collateral and gifts. Nothing is paid for holding it.

Stack — Built, program
Players bond SKR on phones staying face-down. Finishers split 80% of forfeits.

Focus Bond — Built, in the app
A solo bond. The same SKR comes back after a finished shift.

Gift a Rig — Built, program
SOL escrowed for a wallet, or for whoever holds a Seeker's SGT.

SKR fuel — Designed
SKR swapped to SOL in your own clock-in transaction.

20% of forfeits →

a broken bond →

SOL →

SOL →

Bury auction
Forfeited SKR is sold for ORE at a descending price. No oracle.

→ **ORE's own `bury`**
90% burned.
10% distributed by ORE.

An ORE Automation: the recipient's, or your own
The SOL becomes a rig's deposit. Heads Down keeps none of it.

No SKR is emitted and none goes to the team. The Stack and Gift screens are not built in the app. Tested on a fork of live ORE; no SKR flow on mainnet is shown here.

SOURCE docs/SKR.md (flows, §4, §6, §9) · docs/ECONOMICS.md §6 · programs/heads-down/INTERFACE.md §11 · README.md (SKR)

Every rig is an ORE miner, and ORE can recompute our usage from its own events.

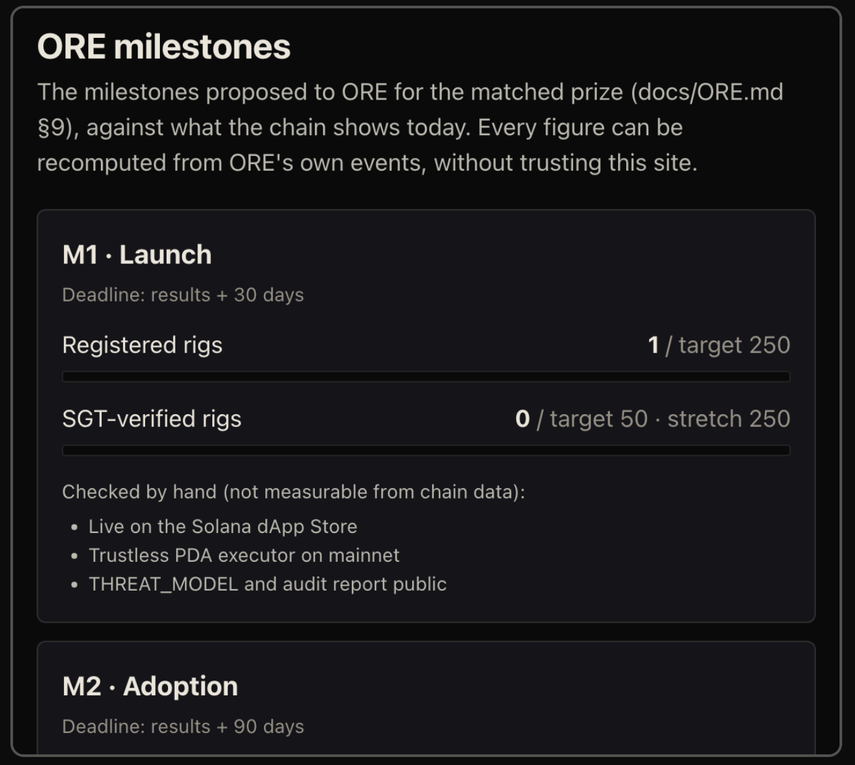
Each rig is your own ORE Automation. Its executor is the Heads Down PDA, on mainnet at By3v…Pkge.

Usage is ORE's `DeployEvents` signed by that PDA, over `ResetEvent.total_miners`.

PROPOSED TO ORE: TARGETS, NONE REACHED

M1 +30 d	dApp Store; 250 rigs, 50 of them SGT-verified
M2 +90 d	1,000 rigs, 300 nightly; 15% of ORE miners per round at night
M3 +180 d	immutable v1; a third-party crank landing digs

Found: ORE stores `max_production_cost`, but `deploy` never reads it. Heads Down enforces its own gate.



The live dashboard, mainnet data, 11 Oct 2026, 20:09 UTC: 1 rig, the founder's.

SOURCE docs/ORE.md §9 · services/indexer/README.md (metric definitions) · deploy/receipts/mainnet/20261010T184715Z-init.json · dashboard-production-b80c.up.railway.app/milestones/

Every key has a written worst case. Mining funds stay in ORE's custody.

KEY	WORST CASE	WHAT BOUNDS IT
Your wallet	Everything it controls. It is the root.	Your wallet's own security; amounts shown before the wallet opens
Rig key, in the phone	Deploys the armed, capped budget into ORE while the gate is open. About 10.5% goes to ORE's fees, none to the attacker.	Wallet-signed caps and expiry; the on-chain gate; Freeze, Unfreeze
Crank	Mining: nothing mines, nothing is lost. A Focus Bond or a Stack seat is lost if no heartbeat lands.	Anyone can run one; only the team's runs today
Registrar	Software keys pass as hardware at capped remote Stack tables.	No custody, no mining power; bond caps
Upgrade key	Today one key, 9DSV…qeeW, no multisig, no delay: the SKR and SOL Heads Down holds; forced deploys up to 25 × the per-square amount a round.	Never an Automation's balance or anyone's ORE. A multisig with a 72 h delay is planned
Team servers	Liveness, privacy, phishing-shaped pushes.	No signing from a push; transactions rebuilt from chain
ORE upstream	ORE owns every Automation and can move funds.	Inherited and stated; layout pins, a circuit breaker

No third-party audit. Our own review and what is still open: docs/SECURITY_REVIEW.md.

SOURCE docs/THREAT_MODEL.md (As built, Summary, K2, K5, K7) · docs/SECURITY_REVIEW.md §3 · deploy/receipts/mainnet/20261010T184116Z-fresh-d67a1a40c2a0.json (upgrade_authority)

OPEN SOURCE

The trust layer is open source, with tests anyone can re-run.

171
`heads_down program`
138 on a fork of live mainnet ORE

214
`crank`
plus 14 on the fork, 1 end to end

116
`registrar`
19 on real Google attestation chains

477
`indexer`
chain data to a public API

1,330
`Android app`
JVM unit tests; 5 more skipped

69
`sgt-verify`
host tests of the Seeker check

48
`dashboard`
every number from chain data

30
`p256-introspect`
the heartbeat's signature check

Security review before deployment: 114 findings, 18 confirmed by a reproducing test. What the first night on mainnet found is written down beside it, with what is still open. On mainnet, the program's bytes equal the build (hash `0154…1b58`).

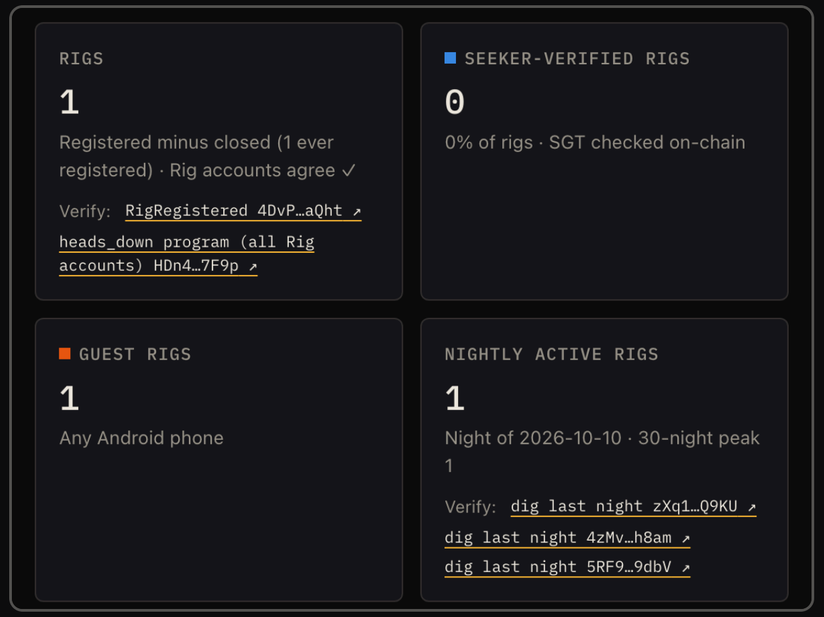
> OPEN TODAY No third-party audit. One key, the founder's, can upgrade the program.

SOURCE docs/SECURITY_REVIEW.md (§3, §5) · README.md (What is built) · crates/sgt-verify/README.md · crates/p256-introspect/README.md · dashboard/test · deploy/receipts/mainnet/20261010T184116Z-fresh-

ON MAINNET

On mainnet since 10 October 2026. One phone has mined its first ORE. No users yet.

Program deployed, slot 455,359,196	4naP…Pae2
Config: 10,000 lamports a dug round, 7,000 to the crank	3R4U…JzUs
Executor PDA, funded with 1,450,240 lamports	By3v…Pkge
Crank, registrar, indexer, dashboard	live, on Railway
Clock-in from the Redmi 14C, signed in Jupiter	4DvP…aQht
First dig, ORE round 435,224	25nB…zovY
Digs landed in that shift (the first was sent too late)	5 of 6 rounds
The pickup, and the BREAK the phone signed	63FZ…yqVb
The same heartbeat sent twice, refused	4EJv…PUpM
Clock-out	3Vfb…uTUm
SOL taken back: 0.00502704 SOL, what the screen said, to the lamport	46NX…DcFF
The shift: 6 rounds dark, 0.005 SOL placed on 20 squares	no ORE won
Since then, on 11 Oct: 4 more shifts, 15 digs in 15 rounds	0.0019 ORE mined



The live dashboard, 11 Oct 2026, 20:09 UTC. The rig is the founder's; 20 rounds dug over three evenings.

DEMO BUILD Its digs run under a raised price ceiling: plan 1.0 SOL per ORE (default 0.53), wallet ceiling 1.2; at most 0.005 SOL a shift, 0.035 a week. One key can upgrade the program. No third-party audit.

SOURCE docs/MAINNET.md (every transaction of the shift, and what it cost) · deploy/receipts/mainnet/ · docs/DEPLOY.md §10.7 · dashboard-production-b80c.up.railway.app

LANDSCAPE

Others mine ORE from a server on a schedule. Heads Down mines only while this phone is down.

	Native Android	Mining gated on the phone	Executor signs only for a phone heartbeat	Mine-or-buy with a published backtest	SGT-verified tier
RefinORE AutoMiner	store app	no	no, a server	not found	not found
Orestack	TWA	no	no, a server	no backtest found	not found
Oreminer	store app	no	no, a server	not found	not found
ore.com	web and PWA	no	no, ORE's own	not found	no
Heads Down	**Kotlin, Compose**	**yes**	**yes**	**yes**	**yes, on a fork**

“Not found” means our review did not find it, not that it cannot exist.

SOURCE docs/research/redteam.md (store listings) · docs/research/skr-and-ore.md (ore.com, RefinORE) · buildplan.md (no phone-gated ORE repo or app found)

WHERE IT STANDS

Deployed on mainnet, worked by a real phone, with the open items written down.

PROVEN ON A FORK OF MAINNET

- The PDA executor, on live ORE
- P-256 heartbeats verified on-chain
- The SGT check
- Stack, Focus Bond, Gift a Rig, the Bury auction

ON MAINNET SINCE 10 OCT 2026

- The program, its bytes equal to the build
- Crank, registrar, indexer, dashboard running
- The Redmi's TEE rig key, attested
- Its first shift, under the raised demo ceiling: 5 digs, a BREAK, a clock-out, SOL back to the lamport; 15 more digs since, and its first 0.0019 ORE

OPEN, AND NEXT

- One upgrade key, then a multisig, then immutable
- No third-party audit
- The Stack and Gift screens; the buy leg
- A whole night, and the morning reveal, on the phone

One builder in Nigeria (WAT), building with AI agents, on one Redmi 14C.

`github.com/OoJae/heads-down` `oojae.github.io/heads-down`

SOURCE README.md · spikes/*/README.md · deploy/receipts/mainnet/ · docs/SECURITY_REVIEW.md §3 (open items) · docs/DEPLOY.md §10.7 (the demo ceiling)